Econ 5725

Quantitative Macro

Wednesdays 4pm-7pm



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Course Description: This class follows the first year PhD macro sequence, Econ 501 and 502. The goal of this course is to (1) introduce the basics tools of modern quantitative macroeconomics, and (2) help students to actually work on quantitative macroeconomics. Applications include models with rich heterogeneity across agents and firms, aggregate fluctuations, and transitional dynamics. This course will require intensive computational work by students. Although this is not a programming class, some knowledge of computer programming will be sueful. If you do not know how to program you will need to work harder. You are free to use whatever programming language you feel comfortable with.

Text(s): We will use several lecture notes available online. However, a good reference is the book Dynamic "General Equilibrium Modeling: Computational Methods and Applications," by Burkhard Heer and Alfred Maussner. I will also use "Applied Computational Economics and Finance," by Mario J. Miranda and Paul L. Fackler and "Economic Dynamics: Theory and Computation," by John Stachurski.

Grade Distribution: Your grade will be based on your grades in the homework assignments and a short final project.

Course Policies:

- **Labs and Assignments**
  - Students are expected to work independently. **Offering** and **accepting** solutions from others is an act of **plagiarism**, which is a serious offense and **all involved parties will be penalized according to the Academic Honesty Policy**. Discussion amongst students is encouraged, but when in doubt, direct your questions to the professor, tutor, or lab assistant.
  - **No late assignments will be accepted under any circumstances**.
- **Attendance and Absences**
  - Attendance is expected at each class.
  - Students are responsible for all missed work, regardless of the reason for absence. It is also the absentee's responsibility to get all missing notes or materials.

Week	Content
CLASS 1. Jan 16	• **Why we do quantitative macroeconomics?** – Lucas (1980), Kydland-Prescott (1996). – **Deterministic Growth Model**. – Extended Path Method. – King and Rebelo (1993).
CLASS 2. Jan 23	• **Stochastic Growth Model**. – Discrete-State-Space Dynamic Programming Problem. – Value function iteration with discretization. – Howards improvement. Other ways to speed-up the algorithm – **Business Cycles**. – Greenwood, Hercowitz, and Huffman (1988). – Computing the model's statistics
CLASS 3. Jan 30	• **Households Heterogeneity in Macroeconomics**. – Why? Krueger, Mitman, and Perri (new handbook chapter). – Macro-micro facts – Kaplan and Violante (2010), Guvenen, Ozkan, and Song (2014) – Data sets. SFC. Kuhn and Rios-Rull (2005).
Feb 6	• **No class.**
CLASS 4. Feb 13	• **The income-fluctuation problem**. – Policy function iteration. 5 alternative methods. – Non-iterative methods. – Accuracy.
CLASS 5. Feb 20	• **Stationary Distribution and GE**. – Monte Carlo and the invariant density function. – Aiyagari. Solving for r^*.

CLASS 6. Feb 27	• **Life-cyle models of consumption, debt, and default**. – Life-cycle models. – Unsecured debt. – Mortgages.
CLASS 7. March 6	• **Sovereign default models**. – Methods. – Calibration. – Divide and conquer.
March 13	• Spring Break
CLASS 8. March 21 (?)	• **Idiosyncratic and Aggregate Risk.** – "MIT shocks". – Transitions. – Krusell & Smith (1998). Boppart, Krusell, & Mitman (2018).
CLASS 9. March 27	• **Firms Heterogeneity in Macroeconomics.** – Cabral & Mata (2003), Midrigan & Xu (2014), and Pugsley, Sedlacek, Sterk (2017). – Hopenhayn (1992). Gomes(2001).
CLASS 10. April 3	• **Firms, Finance, and Default.** – Hennessy and Whited (2007). – Gomes and Schmid (2010), Arellano, Bai, and Zhang (2012).
CLASS 11. April 10	• **Firms and Aggregate Fluctuations.** – Khan, Senga, and Thomas (2013). – Clementi and Palazzo (2016).
CLASS 12. April 17	• **Meet with each group to prepare final project.**
CLASS 13. April 24	• **Final projects presentation.**